LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principles of fuel quantity measurement
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principles of fuel quantity measurement. Practical & Demonstration
5. Specific Objectives	15.1.7.1 Trainee will be able to explain Principles of fuel quantity measurement indicator.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of fuel quantity indicator operation; Safety: Ensure aircraft is grounded and fuel caps secured; Supervision: Instructor monitors trainee observation of float mechanism movement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the fuel quantity indicator?
(b) List two common types of fuel quantity measurement systems.
(c) Why is accurate fuel measurement critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of fuel quantity measurement • Role of float and lever mechanism • Conversion of mechanical movement to electrical signal • Impact of fuel density on measurement accuracy • Effect of aircraft attitude on sensor reading	L & Ds	PPT, IB	15 min
(b)	• Operation of float type gauge mechanism • Signal transmission through fuel tank • Calibration of fuel quantity indicator • Common sources of measurement error • Maintenance requirements for fuel sensors	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of fuel quantity indicator readings • Troubleshooting false low fuel warnings • Verification of fuel quantity during pre-flight • Recording fuel quantity in flight logs • Reporting discrepancies in fuel measurement	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the float mechanism converts fuel level changes into an electrical signal.
(b) Describe two factors that can cause inaccurate fuel quantity readings.
(c) Outline the steps for verifying fuel quantity before engine start.

12. Home Assignment and Follow Up	Prepare a brief note on the constructional details of the float type gauge to be discussed in Lesson Plan 17.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation and constructional details of float type gauge. Practical & Demonstration	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min